J. Derek Lomas, PhD

Tenured Assistant Professor of Positive AI · TU Delft, Industrial Design Engineering, Design Aesthetics Section

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Research Interests

AI & Experience Design, Wellbeing, Human-Centered Artificial Intelligence, Learning Engineering, Quantum Computing, EEG, Music Cognition, Future Studies, Data Science, Cybernetics, Human-Computer Interaction, EdTech, Game Design.

Academic History

2017–present **Industrial Design Engineering, TU Delft.** Tenured Assistant Professor, Design Aesthetics Section. Research on aligning technology with humanity.

2015–2017 **The Design Lab, UC San Diego.** Design Fellow and Postdoctoral Scholar. 2-year appointment under Don Norman. AI-assisted design, AI in education, design for human intelligence.

2009–2014 **HCI Institute, Carnegie Mellon Univers** Interaction. Dissertation: *Optimizing Moti* *Scale Game Design Experiments*. Advisors:

Summer 2008 **Massachusetts Institute of Technology.** Engineering. International Development Design Summit – $10 computers for education in developing contexts

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1999–2003 **Yale University.** B.A. Cognitive Science. Worked with Nick Bostrom.

Entrepreneurial

Playpower Labs (2012–present)

CEO and Chief Learning Scientist. Award-winning learning games used by millions of students. Data-science and game-design contracts with Pearson, Houghton-Mifflin Harcourt, AltSchool, BrainPOP, Scientific Learning, Carnegie Learning. Grants from Gates, MacArthur, and Schmidt foundations.

Playpower Games

Free K–8 math games platform, Common Core / TEKS aligned, with live Kahoot-style quizzes; integrates with Google Classroom, Clever, ClassLink.

EdOptimize / Smart Paper

Computer-vision platform turning hand-completed worksheets into instant feedback. 5M+ students reached across India.

NeuroUX

HIPAA/GDPR-compliant cognitive assessment and Ecological Momentary Assessment (EMA) platform with wearables integrations (Fitbit, Garmin, Apple Watch, ActiGraph).

Source Library

Open digital archive of pre-1700 Hermetic, alchemical, and natural-magic texts. 12,347 books · 11,218 new English translations · 152 languages.

Selected Peer-Reviewed Publications

2026 Lomas JD et al. **Parametric Analysis of Feature-Sp**
Music Imagery and Perception. *IEEE.*

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2025 Liao D, Lomas D, Yu C. **Enhancing the aesthetic appeal of AI-generated physical product designs through LoRA fine-tuning with human feedback.** *AIGC.*

2025 Lomas D, Xue H. **The Harmony of Opposites in Design and Philosophy.** *In: Perspectives on Harmony, ed. Li, Lai, Benitez. Bloomsbury.*

2024 Lomas JD, van der Maden W, Lion G, Bandyopadhyay S, Litowsky Y, Xue H, Desmet P. **Evaluating the alignment of AI with human emotions.** *Advanced Design Research.*

2024 van der Maden W, van Beek E, Halperin BA, Jääskeläinen P, Kang E, Kun P, Lomas JD, van der Burg V. **Death of the Design Researcher? Creating Knowledge Resources for Designers Using Generative AI.** *ACM DIS.*

2024 van der Maden W, Lomas D, Hekkert P. **Developing and evaluating a design method for positive artificial intelligence.** *AI EDAM.*

2023 van der Maden W, Lomas JD, Hekkert P. **A framework for designing AI systems that support community wellbeing.** *Frontiers in Psychology.*

2023 Patel N, Nagpal P, Shah T, Sharma A, Malvi S, Lomas D. **Improving mathematics assessment readability: Do large language models help?** *Journal of Computer Assisted Learning.*

2022 Lin A, Lomas JD. **The Enigma of Mind: A Theory of Evolution and Conscious Experience.** *In: Enigmas – Darwin College Lectures. Cambridge University Press.*

2022 Lomas JD, Lin A, Forster D, Huisman G, Habekost J, Beardow C, Mullen T, Pandey P, Ahmad N, Miyapuram K, van der Maden W, Cross E. **Resonance as a Design Strategy for AI and Social Robots.** *Frontiers in Neurorobotics.*

2022 Lomas D, Xue H. **Harmony in Design: A Synthesis Philosophy, the Sciences, Economics, and Design.**

2022 Shah T, Patel N, Lomas D, Sharma A. **EdOptimize Learning Analytics Platform.** *LAK22.*

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2021 Lomas D, Karac M, Gielen M. **Design Space Cards: Using a Card Deck to Navigate the Design Space of Interactive Play.** *ACM CHI PLAY.*

2019 Lomas D. **How Might Data-Driven Design Help Reduce the Poverty-Achievement Gap?** *Mind, Brain and Education.*

2017 Lomas D, Forlizzi J, Patel N, Shodhan S, Patel K, Poonwala N, Koedinger K. **Is Difficulty Overrated? Investigating the Effects of Choice, Novelty and Suspense on Intrinsic Motivation.** *ACM CHI. Honorable Mention.*

2016 Lomas D, Forlizzi J, Poonwala N, Patel N, Shodhan S, Patel K, Koedinger K, Brunskill E. **Interaction Design as a Multi-Armed Bandit Problem.** *ACM CHI.*

2013 Lomas D, Forlizzi J, Koedinger K. **Optimizing Challenge in an Educational Game Using Large-Scale Design Experiments.** *ACM CHI.*

Full publication list — 75+ papers, 1,500+ citations — at Google Scholar and /research.

Funding

- NWA-ORC Quantum Inspire — PhD student (Caiseal Beardow)
- Quantum Delta "Quantum & Society" with Deborah Nas — Postdoc, 1 day/week personal time
- eHealthjr with Valentijn Visch — Postdoc
- Google AI Gift — €36,000
- KNAW fellowship for a workshop on Resonance — €10,000
- 4TU support on Wellbeing in Engineering Education — ½ PhD
- ErasmusX "AI in University Education" — 1 day/week pe

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Invited Talks (Selected)

2025 **Ozora Festival, Hungary.** Vibes, Resonance and Harmony in the Age of AI

2022	**Experience Research Summit, Salt Lake City.** Design Resonance
2022	**Amplify AI, Amsterdam.** AI for Good Vibes
2022	**Fruit Punch AI, Eindhoven.** AI for Mental Health
2020	**AI Agora, Delft.** AI for Wellbeing
2017	**Mind, Brain and Education, Erice, Sicily.** Large-Scale Learning Systems and the Neuroscience of Poverty
2016	**Latin American Cognitive and Neuroscience School, Argentina.** The Design Challenge of Improving Human Intelligence
2013	**Carnegie Institute of Technology, Pittsburgh.** The Future of Computer-Aided Learning
2010	**Games for Health, Boston.** An Anti-Malaria Game for a $10 Computer
2009	**PopTech, Camden, ME.** Open-Source Games for Global Education
2009	**O'Reilly ETech, San Jose.** Playpower! $12 Computers for Radically Affordable Computer-Aided Learning

Teaching (TU Delft)

- AI Research Methods (graduate)
- Research Methods (graduate & undergraduate, 6× total) — lecture: “Experimental Design for Designers”
- Interactive Technology Design (3×) — lectures: “Big Histo
  Ergonomics”
- Product Use, Understanding and Experience (3×) — lectu
- Entrepreneurial Thinking (5×) — lecture: “AI for Design Ideation”

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- Exploring Interactions (3×); Understanding Humans (2×)

Student Supervision

25 graduated MSc theses, 6 currently supervised, plus PhDs (incl. Willem van der Maden, 2024 – *Designing Positive AI*; Caiseal Beardow, in progress).

Notable entrepreneurial outcomes from supervised theses: Tim Smits (Unpluq, accepted to Techstars); Marco van der Woude (Breeze); Bryan Zaaijer (SenseGlove, leads UX); Ferkan Metin (Envision Glasses, leads UX); Jefta Harwig (Glimp, Philips Entrepreneur Award); Mihovil Karač (Lumies, Shark Tank).

Selected Press

2023	TU Delft Delft Design Stories – “Positive Artificial Intelligence Vibes”
2021	New Scientist, Digital Trends, Lifewire, Inner Detail – coverage of EEG-based song-identification research
2020	UC San Diego Health; News-Medical – “New Mobile Platform Can Address Another Serious Complication of COVID-19”
2014	San Diego Union-Tribune – “How Jargon Is Killing Innovation”; Tech.co – Kaplan/Techstars EdTech Accelerator class
2013	Wall Street Journal – “New Approaches to Teaching Fractions”
2011	KQED MindShift – “Preventing the Sum
2010	NPR; CNN – Playpower / $12 Computer
2009	Wired; The Guardian; Calit2; PopTech So

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2008	ABC News – “Researchers Propose $12 Computer for Developing Countries”

Recognition

- ACM CHI 2017 – Honorable Mention Award
- PopTech Social Innovation Fellow (2009)
- Ashoka Changemakers Innovation Award
- Recognition from Sesame Street, the White House, the MacArthur Foundation, and DARPA for educational-game work

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